Getting Familiar with GRUB

Everything you ever wanted to know about GRUB, but were afraid to ask, is incorporated in this very informative article.

Whenever a computer is switched on, the boot loader is the first software program to run on it. The boot loader is responsible for the loading of the kernel software, which in turn loads the operating system. GNU GRUB is a multi-boot boot loader and is one of the most used. Most of the popular operating systems have GNU GRUB because of its user-friendly design and support for multiple boots.

History of GRUB

It all started in 1995, when Erich Boleyn was trying to boot the GNU Hurd with the GNU Mach (earlier known as the University of Utah's Mach 4 microkernel). The aim of developing GRUB was mainly to offer support for compatibility on different machines and operating systems. The idea of modifying the FreeBSD boot loader was dropped when Boleyn realised that it was easier to develop a boot loader rather than modify the current one. GRUB had a lot of features added by Boleyn, which makes it stand out from all the existing boot loaders.

Later, in 1999, GRUB was adopted as an official GNU package by Gordon Matzigkeit and Yoshinori K. Okuji. The latest sources were made available through Anonymous CVS for the development of GRUB.

In later years, GRUB was further developed and more features were added. But it soon became obvious that the design of GRUB was incompatible with the new features that were planned for it.

Around 2002, Okuji started rewriting GRUB to make it cleaner, safer and more powerful than the previous version, which was known as Preliminary Universal Programming Architecture (PUPA for GNU GRUB). PUPA was eventually renamed GRUB 2, and the original GRUB was named GRUB Legacy. The last stable release for GRUB Legacy came out in 2005.

GRUB 2 was widely accepted by the computer science community. Most of the Linux distributions started using GRUB 2 and by the end of 2009, multiple major distributions were installing it by default.

The booting process

Ever thought about what really happens when you switch on a computer and how the operating system is loaded? Booting can be defined as a series of processes that help to start a computer and put it into a state of readiness for any operation by the user. Booting is said to be complete when the normal, operative, runtime environment is attained. Booting of a computer can be explained in several phases.

The BIOS phase: BIOS (Basic Input Output System) is a firmware interface. The BIOS software is stored on a ROM chip on the motherboard and is the first software run by a PC when powered on. It carries out the process called POST (Power on Self-Test) which checks, identifies and initialises the hardware to find that every device has the right amount of power, and performs a memory test to see whether the memory is corrupted. The BIOS cannot load the kernel directly because it is unable to access enough of the large hard disk space to load the kernel code. Instead,